

Phase 1 — Fundamental Blast Vibration Theory

1. Nicholls et al. (1971) — USBM / Holmberg–Persson Empirical Formula

What it is

The foundational empirical blast vibration prediction approach used across mining and blasting engineering. Introduces scaled distance–based PPV prediction.

Why you should read it

This is your baseline model. Your ML and ensemble methods must outperform this classical empirical formulation.

Read for

- Scaled distance concept
- Attenuation laws
- Site constants
- Classical PPV prediction philosophy

Direct Link

<https://www.osmre.gov/resources/blasting/docs/USBM/ri8507.pdf>

2. Investigation of Effects of Blasting Design Parameters and Rock Properties on Blast-Induced Ground Vibrations

What it is

A triaxial blast vibration study analyzing longitudinal, transverse, and vertical PPV components separately using regression methods.

Why you should read it

This is one of the closest papers to your work because it studies component-wise vibration behavior, but without machine learning.

Read for

- Component-level PPV analysis
- V/L/T vibration measurements
- Rock property influence
- Monitoring methodology

Direct Link

https://www.researchgate.net/publication/271915137_Investigation_of_effects_of_blasting_design_parameters_and_rock_properties_on_blast-induced_ground_vibrations

3. Component PPV and PVS Correlation — Sijaying Open-Pit Mine

What it is

A directional vibration propagation study showing anisotropic amplification depending on excavation direction.

Why you should read it

This gives direct evidence for the directional amplification phenomenon your paper is centered on.

Read for

- Directional amplification
- Excavation-direction effects
- Component correlations
- Anisotropic vibration propagation

Direct Link

https://www.researchgate.net/figure/Component-PPV-and-PVS-correlation_tbl4_271274704

4. Blasting Vibration Analyses of Open-Pit Mine Slope (2026)

What it is

A modern investigation discussing limitations of current PPV-only analyses and the need for richer vibration descriptors.

Why you should read it

Useful for writing your research gap and motivation section.

Read for

- Spectral analysis
- PPV vectors
- Frequency-domain vibration analysis
- Current limitations in blasting vibration research

Direct Link

<https://www.sciencedirect.com/science/article/pii/S1674775526000119>

Phase 2 — Ensemble Machine Learning for PPV

5. Ensemble XGBoost + ANN for PPV — Scientific Reports (2023)

What it is

A stacked ensemble framework combining ANN and XGBoost for high-accuracy PPV prediction.

Why you should read it

This is your methodological benchmark for ensemble learning.

Read for

- Ensemble stacking
- XGBoost pipelines
- ANN integration
- Feature importance
- Performance evaluation

Direct Link

<https://www.nature.com/articles/s41598-023-33796-7>

6. RVM Dual-Kernel Approach — Scientific Reports (2024)

What it is

A machine learning PPV prediction study explicitly defining triaxial PPV components.

Why you should read it

Useful mainly for formal definitions and terminology related to V/L/T components.

Read for

- Triaxial PPV definitions
- Component terminology
- Vibration vector decomposition

Direct Link

<https://www.nature.com/articles/s41598-024-70939-w>

7. SVR + Optimization for Open-Pit PPV

What it is

An ML-based PPV prediction paper using support vector regression and optimization methods.

Why you should read it

Strong background reference explaining why PPV prediction matters in mining environments.

Read for

- Structural vibration impacts
- PPV importance
- ML vs empirical approaches
- Ground vibration propagation

Direct Link

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6983179/>

Phase 3 — SHAP and Explainable AI

8. CCO-XGBoost + SHAP — Panzhihua Iron Mine (2025)

What it is

An explainable XGBoost framework using SHAP to analyze PPV prediction drivers.

Why you should read it

This paper applies SHAP to resultant PPV. Your work extends this to component-wise SHAP analysis.

Read for

- SHAP methodology
- Feature contribution analysis
- Distance attenuation effects
- Explainable ML workflow

Direct Link

<https://www.mdpi.com/2227-7390/14/4/596>

9. CatBoost + SHAP + GAM — Mirador Copper Mine (2025)

What it is

A modern explainable ML study integrating CatBoost, SHAP, GAM, and Optuna optimization.

Why you should read it

One of the newest and strongest interpretability-focused PPV studies.

Read for

- CatBoost
- Optuna tuning
- SHAP interpretation
- GAM-based explainability

Direct Link

<https://doi.org/10.1177/10775463251389029>

10. RUN-XGBoost + SHAP — Panzhihua Mine (2024)

What it is

An optimized XGBoost framework outperforming empirical PPV equations.

Why you should read it

Useful for arguing why nonlinear ML methods outperform classical blasting formulas.

Read for

- ML vs empirical comparison
- Nonlinear prediction
- SHAP contribution analysis

Direct Link

<https://pubmed.ncbi.nlm.nih.gov/38689981/>

11. IHO-VMD-CatBoost + SHAP — Oxford JCDE (2025)

What it is

A hybrid vibration prediction model with detailed SHAP interpretation at global and local levels.

Why you should read it

Excellent reference for structuring your SHAP results section.

Read for

- Local SHAP analysis
- Global SHAP analysis
- Feature interactions
- Interpretability structure

Direct Link

<https://academic.oup.com/jcde/article/12/2/154/7959003>

Phase 4 — Supporting Hybrid Model Papers

12. CB-BOA Hybrid Model (2025)

What it is

A hybrid CatBoost optimization framework with uncertainty analysis.

Why you should read it

Good support paper for discussing hybrid optimization methods and reliability analysis.

Read for

- Hybrid ML
- Optimization
- Prediction intervals
- Uncertainty analysis

Direct Link

<https://www.sciencedirect.com/science/article/pii/S2214914725002028>

13. Hybrid RF Approach — Wujiata Coal Mine

What it is

A review-style machine learning PPV prediction paper discussing influential blasting variables.

Why you should read it

Good background literature for your introduction and related work section.

Read for

- Influential PPV variables
- Geological effects
- Empirical vs ML approaches

Direct Link

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11680976/>